

HTML Fundamentals

Course Information

Course Name: HTML Fundamentals

Level: Beginner

Prerequisites: Basic computer knowledge

Estimated Duration: 8–10 Hours

Course Description

HTML (HyperText Markup Language) is the standard markup language used to create and structure content on the web. In this course, students will learn how to create web pages using HTML, including headings, paragraphs, links, images, lists, tables, forms, semantic elements, and multimedia. By the end of the course, students will be able to create a complete structured webpage.

Learning Outcomes

After completing this course, students will be able to:

- Explain what HTML is and how it works.
 - Create a basic HTML document.
 - Use HTML elements and attributes.
 - Add headings, paragraphs, links, and images.
 - Create ordered and unordered lists.
 - Create tables.
 - Build forms and collect user input.
 - Embed audio and video.
 - Use semantic HTML elements.
 - Create a structured webpage.
 - Apply basic HTML accessibility practices.
-

Lesson 1: Introduction to HTML

Learning Objectives

After this lesson, students will be able to:

- Define HTML.
- Explain the purpose of HTML.
- Understand how HTML works with web browsers.

- Create a basic HTML document.

What is HTML?

HTML stands for **HyperText Markup Language**.

It is used to structure content on web pages.

HTML can define:

- Headings
- Paragraphs
- Images
- Links
- Lists
- Tables
- Forms
- Videos
- Audio

HTML is a **markup language**, not a programming language.

HTML and Web Browsers

A browser such as Chrome, Firefox, or Edge reads HTML code and converts it into a visual webpage.

For example:

```
<h1>Hello World</h1>
```

The browser displays:

Hello World

Basic HTML Document

Every HTML page normally starts with a document structure like this:

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Page</title>
  </head>
  <body>

    <h1>Hello World</h1>
```

```
<p>This is my first webpage.</p>  
  
</body>  
</html>
```

Important Parts

DOCTYPE

```
<!DOCTYPE html>
```

Tells the browser that the document uses HTML5.

html

```
<html>  
</html>
```

Contains the complete HTML document.

head

```
<head>  
</head>
```

Contains information about the webpage.

title

```
<title>My Website</title>
```

Defines the title shown in the browser tab.

body

```
<body>  
</body>
```

Contains the visible webpage content.

Lesson 2: HTML Elements and Attributes

HTML Elements

An HTML element normally consists of an opening tag, content, and a closing tag.

```
<p>Hello World</p>
```

Here:

- `<p>` is the opening tag.
- `Hello World` is the content.
- `</p>` is the closing tag.

HTML Tags

Examples include:

```
<h1>Heading</h1>  
<p>Paragraph</p>  
<div>Container</div>
```

Some elements do not require a closing tag.

For example:

```

```

Attributes

Attributes provide additional information about an HTML element.

Example:

```
<a href="https://example.com">Visit Website</a>
```

The `href` attribute specifies the destination of the link.

Another example:

```

```

The attributes are:

- `src`
- `alt`

Lesson 3: Headings and Paragraphs

Headings

HTML provides six levels of headings.

```
<h1>Main Heading</h1>
<h2>Section Heading</h2>
<h3>Subsection Heading</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

`<h1>` represents the highest-level heading.

Paragraphs

Use the `<p>` element for paragraphs.

```
<p>
HTML is used to structure content on the web.
</p>
```

Line Break

The `<br>` element creates a line break.

```
<p>Hello<br>World</p>
```

Horizontal Line

The `<hr>` element creates a thematic horizontal line.

```
<hr>
```

Lesson 4: Text Formatting

HTML provides elements for giving meaning or emphasis to text.

Bold Text

```
<strong>Important text</strong>
```

Italic or Emphasized Text

```
<em>Emphasized text</em>
```

Highlighted Text

```
<mark>Highlighted text</mark>
```

Deleted Text

```
<del>Old price</del>
```

Subscript

```
H<sub>2</sub>O
```

Superscript

```
x<sup>2</sup>
```

Lesson 5: Links

Creating a Link

Use the `<a>` element.

```
<a href="https://www.google.com">
  Visit Google
</a>
```

Open Link in New Tab

```
<a href="https://www.google.com" target="_blank">
  Visit Google
</a>
```

Link to Another Page

```
<a href="about.html">About Us</a>
```

Email Link

```
<a href="mailto:example@email.com">
  Send Email
</a>
```

Lesson 6: Images

Adding an Image

Use the `<img>` element.

```

```

Important attributes:

- `src` specifies the image location.
- `alt` provides alternative text.

Image Dimensions

```

```

Why is alt Important?

The `alt` attribute helps:

- Screen readers
- Accessibility
- Users when an image cannot load

Example:

```

```

Lesson 7: Lists

HTML supports ordered and unordered lists.

Unordered List

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Output:

- HTML
 - CSS
 - JavaScript
-

Ordered List

```
<ol>
  <li>Open the browser</li>
  <li>Open the website</li>
  <li>Login</li>
</ol>
```

Nested Lists

Lists can contain other lists.

```
<ul>
  <li>Programming
    <ul>
      <li>JavaScript</li>
      <li>Python</li>
    </ul>
  </li>
</ul>
```

Lesson 8: Tables

HTML tables are used to display structured data.

Basic Table

```
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>

  <tr>
    <td>Ali</td>
    <td>20</td>
  </tr>

  <tr>
    <td>Ahmed</td>
    <td>22</td>
  </tr>
</table>
```

Important Table Elements

table

Defines the table.

tr

Defines a table row.

th

Defines a table header cell.

td

Defines a table data cell.

Table Caption

```
<table>
  <caption>Student Information</caption>

  <tr>
    <th>Name</th>
    <th>Marks</th>
```

```
</tr>

<tr>
  <td>Ali</td>
  <td>85</td>
</tr>
</table>
```

Lesson 9: Forms

HTML forms are used to collect information from users.

Basic Form

```
<form>
  <label>Name:</label>
  <input type="text">

  <button type="submit">
    Submit
  </button>
</form>
```

Input Types

HTML provides many input types.

```
<input type="text">

<input type="email">

<input type="password">

<input type="number">

<input type="date">

<input type="file">

<input type="checkbox">

<input type="radio">
```

Labels

Labels help users understand what information should be entered.

```
<label for="email">
  Email:
</label>

<input
  type="email"
  id="email"
>
```

Textarea

Used for multi-line text.

```
<textarea>
</textarea>
```

Select Dropdown

```
<select>
  <option>Pakistan</option>
  <option>India</option>
  <option>Bangladesh</option>
</select>
```

Lesson 10: Multimedia

HTML supports multimedia content.

Audio

```
<audio controls>
  <source src="music.mp3" type="audio/mpeg">
</audio>
```

The `controls` attribute provides playback controls.

Video

```
<video controls width="500">
  <source src="video.mp4" type="video/mp4">
</video>
```

YouTube Video

Videos can also be embedded using an iframe.

```
<iframe
  src="https://www.youtube.com/embed/VIDEO_ID"
  width="560"
  height="315">
</iframe>
```

Lesson 11: HTML Semantic Elements

Semantic elements describe the meaning of the content.

Common semantic elements include:

```
<header>
<nav>
<main>
<section>
<article>
<aside>
<footer>
```

Example

```
<header>
  <h1>My Website</h1>
</header>

<nav>
  <a href="index.html">Home</a>
  <a href="about.html">About</a>
```

```
</nav>

<main>
  <section>
    <h2>About Us</h2>
    <p>Welcome to our website.</p>
  </section>
</main>

<footer>
  <p>Copyright 2026</p>
</footer>
```

Why Use Semantic HTML?

Semantic HTML:

- Improves accessibility.
- Makes code easier to understand.
- Helps search engines understand webpage structure.
- Makes websites easier to maintain.

Lesson 12: Div and Span

div

`div` is a general-purpose block-level container.

```
<div>
  <h2>Student Information</h2>
  <p>Name: Ali</p>
</div>
```

span

`span` is an inline container.

```
<p>
  This is <span>important</span> information.
</p>
```

Lesson 13: HTML Comments

Comments are ignored by the browser.

```
<!-- This is an HTML comment -->
```

Comments can be used to explain code.

Example:

```
<!-- Student information section -->
<section>
  <h2>Students</h2>
</section>
```

Lesson 14: HTML Entities

Some characters have special meanings in HTML.

For example:

```
&lt;
&gt;
&amp;
&quot;
&nbsp;
```

Examples:

```
<p>5 &lt; 10</p>
```

```
<p>Tom &amp; Jerry</p>
```

Lesson 15: HTML Accessibility

Accessibility means making websites usable by as many people as possible, including people who use assistive technologies.

Alternative Text

Always provide meaningful `alt` text for informative images.

```

```

Labels for Forms

Use labels with form controls.

```
<label for="username">  
  Username  
</label>  
  
<input  
  type="text"  
  id="username"  
>
```

Proper Heading Structure

Use headings in a logical order.

```
<h1>HTML Course</h1>  
  
<h2>HTML Basics</h2>  
  
<h3>HTML Elements</h3>
```

Lesson 16: Complete Webpage Example

The following example combines many concepts from the course.

```
<!DOCTYPE html>  
<html lang="en">  
  
<head>
```



```
<meta charset="UTF-8">

<meta
  name="viewport"
  content="width=device-width, initial-scale=1.0"
>

<title>Student Profile</title>
</head>

<body>

  <header>
    <h1>Student Profile</h1>

    <nav>
      <a href="#about">About</a>
      <a href="#skills">Skills</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <main>

    <section id="about">

      <h2>About Me</h2>

      

      <p>
        Hello! I am a student learning web development.
      </p>

    </section>

    <section id="skills">

      <h2>My Skills</h2>

      <ul>
        <li>HTML</li>
        <li>CSS</li>
        <li>JavaScript</li>
      </ul>

    </section>
```

```
<section id="contact">

    <h2>Contact Me</h2>

    <form>

        <label for="name">
            Name:
        </label>

        <input
            type="text"
            id="name"
            required
        >

        <br><br>

        <label for="email">
            Email:
        </label>

        <input
            type="email"
            id="email"
            required
        >

        <br><br>

        <button type="submit">
            Send
        </button>

    </form>

</section>

</main>

<footer>
    <p>© 2026 Student Profile</p>
</footer>

</body>

</html>
```

Mini Project

Personal Portfolio Website

Create a personal portfolio website using HTML.

Requirements

The website should contain:

1. Header
2. Navigation menu
3. About section
4. Profile image
5. Skills section
6. Education section
7. Projects section
8. Contact form
9. Footer

Concepts to Use

The project should demonstrate:

- HTML document structure
 - Headings
 - Paragraphs
 - Links
 - Images
 - Lists
 - Tables
 - Forms
 - Semantic elements
 - Attributes
-

Practice Exercises

Exercise 1

Create a webpage containing:

- One heading
- Three paragraphs
- One image
- One link

Exercise 2

Create a student table containing:

- Student name
- Roll number
- Course
- Marks

Exercise 3

Create a registration form containing:

- Name
- Email
- Password
- Date of birth
- Gender
- Country
- Submit button

Exercise 4

Create a webpage containing:

- Header
- Navigation
- Main content
- Three sections
- Footer

Final Quiz

1. What does HTML stand for?
2. Is HTML a programming language?
3. What is the purpose of `<!DOCTYPE html>`?
4. What is an HTML element?
5. What is an HTML attribute?
6. What is the purpose of the `href` attribute?
7. What is the purpose of the `alt` attribute?
8. What is the difference between `<ol>` and `<ul>`?
9. What are `<tr>`, `<th>`, and `<td>` used for?
10. What is the purpose of an HTML form?
11. What is the difference between `div` and `span`?
12. What are semantic HTML elements?
13. Why is accessibility important?
14. What is the purpose of the `<label>` element?
15. What is the purpose of the `<iframe>` element?

Course Summary

After completing this course, students should understand the fundamentals of HTML and be able to create structured webpages.

The major concepts covered are:

- HTML document structure
- Elements and attributes
- Headings and paragraphs
- Text formatting
- Links
- Images
- Lists
- Tables
- Forms
- Multimedia
- Semantic HTML
- Div and span
- Comments
- HTML entities
- Accessibility
- Complete webpage structure

Next Step

After completing HTML Fundamentals, students should learn **CSS** to style and design their HTML webpages.

After CSS, they can continue with **JavaScript** to add interactivity and dynamic behavior.